SKILL BUILDERS · UNIT 3 · SCORE 3

Equations & Inequalities: Practice Set

NC.8.EE.7

ANSWER KEY

Do one problem at a time. Show your steps. Circle the letter or write the number.

1 Score 3 NC.8.EE.7

Solve: $x - 7 = 10$

Answer: **17**

1. $x - 7 = 10$ → Add 7
   $+7 \quad +7$
2. $x = 17$

2 Score 3 NC.8.EE.7

Solve: $3x = 18$

Answer: **6**

1. $\frac{3x}{3} = \frac{18}{3}$ → Divide by 3
2. $x = 6$

3 Score 3 NC.8.EE.7

Solve: $2x + 5 = 19$

Answer: **7**

1. $2x + 5 = 19$ → Subtract 5
   $-5 \quad -5$
2. $\frac{2x}{2} = \frac{14}{2}$ → Divide by 2
3. $x = 7$

4 Score 3 NC.8.EE.7

Solve: $4x - 3 = 25$

Answer: **7**

1. $4x - 3 = 25$ → Add 3
   $+3 \quad +3$
2. $\frac{4x}{4} = \frac{28}{4}$ → Divide by 4
3. $x = 7$

5 Score 3 NC.8.EE.7

Solve: $4(x + 1) = 24$

Answer: 5

1. $4(x + 1) = 24$ ↓ Distribute 4
2. $4x + 4 = 24$, $-4 \quad -4$ ↓ Subtract 4
3. $\frac{4x}{4} = \frac{20}{4}$ ↓ Divide by 4
4. $x = 5$

6 Score 3 NC.8.EE.7

Solve: $5x - 2 = 3x + 8$

Answer: 5

1. $5x - 2 = 3x + 8$, $-3x \quad -3x$ ↓ Subtract 3x
2. $2x - 2 = 8$, $+2 \quad +2$ ↓ Add 2
3. $\frac{2x}{2} = \frac{10}{2}$ ↓ Divide by 2
4. $x = 5$

7 Score 3 NC.8.EE.7

Solve: $3(x - 2) + x = 10$

Answer: 4

1. $3(x - 2) + x = 10$ ↓ Distribute 3
2. $3x - 6 + x = 10$ ↓ Combine like terms
3. $4x - 6 = 10$, $+6 \quad +6$ ↓ Add 6
4. $\frac{4x}{4} = \frac{16}{4}$ ↓ Divide by 4
5. $x = 4$

8 Score 3 NC.8.EE.7

How many solutions does $2x + 6 = 2x + 1$ have?

(A) **No solution** (B) One solution

(C) Infinitely many solutions

1. $2x + 6 = 2x + 1$, $-2x \quad -2x$ ↓ Subtract 2x
2. $6 = 1$

1 $6 = 1$ is false → no solution

9 Score 3 NC.8.EE.7

Solve: $-3x \leq 12$

(A) $x \geq -4$ (B) $x \geq 4$

(C) $x \leq -4$ (D) $x \leq -36$

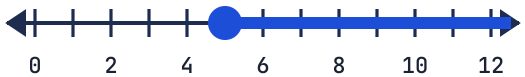
1 $-3x \leq 12$

2 $x \geq -4$ ↓ Divide by −3 and FLIP the sign

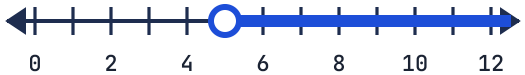
10 Score 3 NC.8.EE.7

Which graph shows $x + 4 > 9$?

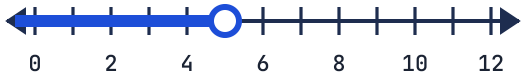
(A) 5



(B) 5



(C) 5



(D) 9



1 $x + 4 > 9$

2 $x > 5$ ↓ Subtract 4

3 open circle at 5, shade right

11 Score 3 NC.8.EE.7

Find the mistake. Jordan solved $2(x + 3) = 14$:

Line 1: $2x + 3 = 14$

Line 2: $2x = 11$

Line 3: $x = 5.5$

Where is the mistake?

A **Line 1: the 2 was not multiplied by 3. It should be 2x + 6 = 14.**

B Line 2: 14 − 3 is not 11.

C There is no mistake.

D Line 3: 11 ÷ 2 is not 5.5.

1. $2(x + 3) = 14$ ↓ Distribute 2 to BOTH terms
2. $2x + 6 = 14$ ↓ Subtract 6
   $-6 \quad -6$
3. $\frac{2x}{2} = \frac{8}{2}$ ↓ Divide by 2
4. $x = 4$

12 Score 3 NC.8.EE.7

Five more than twice a number is 17. The equation is $2x + 5 = 17$. What is the number?

Answer: **6**

1. $2x + 5 = 17$ ↓ Subtract 5
   $-5 \quad -5$
2. $\frac{2x}{2} = \frac{12}{2}$ ↓ Divide by 2
3. $x = 6$